I.

Reference solution: is an arbitrary constant).

Reference solution:

Reference solution:

Reference solution:

; is electron charge magnitude, Boltzmann's constant, and

is absolute temperature; is the reverse saturation current of the junction.

II.

Reference solution:

. Scaling position and time as gives velocity and acceleration:

Both the original and scaled particle motion obey Newton's equations:

Using and becomes

Comparison with gives

. For gravity, . If the orbital size scales as

its period scales as